MILITARY SPECIFICATION

DIRECTION FINDER SET AN/ARN-89( )

1. SCOPE

1.1 This specification covers the following units of one type of automatic direction finder airborne set which operates throughout the frequency range of 100 to 3000 KHz and is designated as Direction Finder Set AN/ARN-89( ), (see 6.1 and 6.5):

Amplifier, Impedance Matching	AM-4859( )/ARN-89
Antenna	AS-2108( )/ARN-89
Control, Radio Set	C-7392( )/ARN-89
Receiver, Radio	R-1496( )/ARN-89

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS:

FEDERAL

PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple Wall
PPP-C-850	Cushioning Material, Polystrene Expanded, Resilient (for packaging uses).
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-S-760	Strapping, Nonmetallic, (and Connectors).
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-P-116	Preservation, Methods of
MIL-B-117	Bag and Sleeve, Interior Packaging
MIL-B-131	Barrier Material, Water Vaporproofed, Flexible.

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MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-8835	Indicator, Humidity, Card, Chemically Impregnated
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Signal Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-B-43291	Boxes; Fiberboard; Corrugated, Double-Wall, Weather Resistant
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads; (40 in. x 48 in. 4-Way Partial and 4-Way Pallets).
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Tests Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods

DRAWINGS

ELECTRONIC COMMAND

SC-DL-596907	Receiver, Radio R-1496( )/ARN-89
SC-DL-597300	Control, Radio Set C-7392( )/ARN-89
SC-DL-597450	Amplifier, Impedance Matching AM-4859( )/ARN-89
SC-DL-597502	Antenna AS-2108( )/ARN-89

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(Copies of specifications standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.).

3. REQUIREMENTS

3.1 Description - The Direction Finder Set AN/ARN-89( ) is a lightweight automatic direction finder (ADF) set capable of receiving modulated or unmodulated signals by use of either a sense or a loop antenna over the frequency range of 100 to 3000 KHz and determining the source of transmission automatically and continuously seeking the correct bearing of the transmission to which the receiver is tuned. The bearing angle is shown by the use of a remote bearing indicator ID-1351/AR. The sense antenna is a suitable non-directional antenna with an effective height between 0.05 and 0.5 meters with a capacitance of 40 to 100 PF. The ADF provides means for (I) continuous and automatic indication in degrees, corrected for radio compass deviation, of the direction of arrival of modulated or unmodulated signals, and in simultaneous aural reception of such signals, by use of a loop and a sense antenna, (II) aural reception of modulated or unmodulated signals by use of either a sense or a loop antenna, with means for electrically rotating the loop for aural-null determination of bilateral bearing, (III) complete control from a single position and bearing indication at that position. The equipment operates from an aircraft power supply having a nominal voltage of 28 volts dc.

3.2 Construction - The units on order shall be constructed in accordance with the following drawings and data lists as specified:

SC-DL-596907	Receiver, Radio R-1496( )/ARN-89
SC-DL-597300	Control, Radio Set C-7392( )/ARN-89
SC-DL-597450	Amplifier, Impedance Matching AM-4859( )/ARN-89
SC-DL-597502	Antenna AS-2108( )/ARN-89

3.3 Preproduction Samples - The contractor shall furnish preproduction samples of equipment on order as required by the invitation for bids and contract. (See 4.3 & 6.2).

3.4 Cleaning -

3.4.1 Parts - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.5 Controls - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 Finish, protective (See 4.4) - The equipment shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type I surfaces shall be lusterless black conforming to Specification MIL-E-14072, except the final color on Type I surfaces for the AS-2108( )/ARN-89 shall be in accordance with the equipment drawings.'

3.7 Securing of parts -

3.7.1 General - Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7.2 Securing of parts by threaded fasteners - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250¼F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.8 Marking (See 4.4)

3.8.1 General - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be in accordance with the equipment drawings.

3.8.2 Visibility - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.8.3 Serial Numbers - The unit(s) listed in paragraph 1.1 shall be serial numbered.

3.9 Soldering -

3.9.1 Solder - Solder shall conform to the equipment drawings.

3.9.2 Acid or acid salts - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

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3.9.3 Process- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring and cabling - Wiring and cabling shall be neat and sturdy.

3.10.1 Cabling - Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance.

3.10.2 Slack - Wires and cables shall be as short as practicable except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected, except where uninsulated wires are used.

3.10.3 Protection - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.10.4 Support - Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (I) during and after subjection of the equipment to specified service conditions or (II) after service or repair of the equipment in a normal manner.

3.10.5 Clearance - Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.6 Splicing- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

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3.10.7 Connections, general - Before being soldered to terminal lugs or fixed terminals, wire shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to complete electrical connections.

3.11 Printed wiring assemblies (See 4.4) - Printed wiring assemblies shall be processed for compliance with the requirements specified on the equipment drawings and MIL-P-55110 in order not to affect life and serviceability of the assemblies.

3.12 Semiconductors (transistors and diodes) (See 4.4)- These electron devices (semiconductors and diodes) shall conform to Specification MIL-S-19500 and the equipment drawings.

3.13 Tropicalization of materials - Materials shall be treated as follows:

3.13.1 Treating materials- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.13.2 Toxicity - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.13.3 Flexibility - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.13.4 Statement of treatment - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2).

3.14 Service conditions (See 4.8) - The units shall meet the service conditions as specified in Specification MIL-E-5400 for Class I equipment except as otherwise specified herein, and shall meet the specified requirements herein, after a warm-up period not to exceed 60 seconds.

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3.14.1 Temperature (See 4.8.1)-

3.14.1.1 Operating (See 4.8.1.1) - Operable conforming to limits specified from -25°F to +160°F for continuous operation. Operable from -65°F to -25°F and no measurement shall deteriorate by more than a factor of two. The equipment shall be considered Class 1 except when otherwise specified.

3.14.1.2 Non-Operating (See 4.8.1.2) - Minimum exposure of 24 hours at -65°F and four (4) hours at maximum of +160°F.

3.14.2 Altitude (See 4.8.2) - Operational during the required temperature altitude steps up to 50,000 feet above sea level.

3.14.2.1 Temperature-Altitude Combination (See 4.8.2.1) - The unit(s) shall operate under temperature-altitude combinations required by MIL-E-5400, Class I equipment.

3.14.3 Relative humidity (See 4.8.3)- Up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. This test shall be performed over a 10 day cycle.

3.14.4 Sand and dust (See 4.8.4)- The unit(s) shall withstand exposure to sand and dust particles as encountered in desert areas.

3.14.5 Vibration (See 4.8.5) - The unit(s) shall meet the vibration curve III for helicopter equipment.

3.14.6 Shock (See 4.8.6)-

3.14.6.1 Bench handling (See 4.8.6.1)- The unit(s) shall be capable of full specification performance after subjection to the required numbers of drops.

3.14.6.2 Equipment (See 4.8.6.2)- The unit(s) shall meet the equipment shock test. The unit(s) shall be capable of meeting full specification performance after the shock test.

3.14.7 Fungus (See 4.8.7) The unit(s) shall show no evidence of viable fungus or corrosion after being subjected to 28 days of fungus test.

3.14.8 Salt fog (See 4.8.8)- The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.14.9 Explosive proof (See 4.8.9)- The unit(s) shall not cause ignition of any ambient-explosive-gaseous mixture with air when operative in such an atmosphere.

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3.14.10 Acceleration (See 4.8.10)- The unit(s) shall operate within specified limits during application of 6g steady state acceleration for one (1) minute.

3.14.11 Warm-up equipment (See 4.8.11)- The requirements specified for audio output shall be met within 30 seconds after power is applied to the unit(s) at room ambient conditions.

3.14.12 Reliability (See 4.8.12)- The equipment shall have a reliability sufficient to provide a mean-time-between-failures of 660 hours, when the units meet all tests as specified.

3.15 Electrical Requirements (See 4.9)-

3.15.1 Power Supply (See 4.9.1)-

3.15.1.1 Power supply primary (See 4.9.1.1)- Except as specified herein the equipment shall be capable of operating from a 27.5 volts dc aircraft power as defined in MIL-STD-704, Category B, and the total input current shall not exceed 1.3 amp with 50 mw of audio output while the goniometer is being rotated, using the internal 400 Hz supply.

3.15.1.2 Power supply, 400 Hz (See 4.9.1.2) - The equipment shall provide a 26 volt ± 20%, 400 Hz +10-30% output voltage. The selection of this internal 400 Hz supply or external 400 Hz supply shall be accomplished at the aircraft interface connector.

3.15.2 ANT, Mode (See 4.9.2)-

3.15.2.1 Sensitivity, ant (See 4.9.2.1)- The antenna sensitivity shall be at least, 10.0 microvolts from 110 kHz up to, but not including, 200 kHz, and 5.0 microvolts from 200 kHz up to 3000 kHz for a 6 db signal-plus-noise-to-noise ratio at the audio output when receiving a signal modulated 30 percent by 1000 Hz. With these inputs the gain shall be such that the audio output will produce 50 milliwatts of signal across a 150 ohm load. Sensitivity degradation shall not exceed 6 db, with 50 milliwatt output, during subjection to any service condition.

3.15.2.2 Sensitivity, CW, ant (See 4.9.2.2)- When RF inputs of 3.2 microvolts from 100 kHz up to but not including 200 kHz and 1.6 microvolts from 200 kHz up to 3000 kHz are applied at the specified frequencies and the audio output is adjusted for 10 milliwatts across 150 ohm load the audio output shall be less than 2.5 milliwatts when the input signal is removed.

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3.15.2.3 I.F. rejection, ant (See 4.9.2.3)- The intermediate frequency rejection shall be at least 80 db down at any frequency to which the equipment is tuned.

3.15.2.4 Image rejection, ant (See 4.9.2.4)- The image rejection ratio shall be at least 80 db down at any frequency within the tuning range of the equipment.

3.15.2.5 Selectivity, ant (See 4.9.2.5)-

a. 4 kHz ± 1.0 kHz bandwidth at 6 db points.

b. 12 kHz maximum bandwidth at 60 db.

3.15.2.6 Cross modulation, ant (See 4.9.2.6)- With the "AUDIO" control set to give 20 milliwatts audio output at 1000 Hz with a 50 microvolt 30 percent modulated R.F. signal, the difference between the frequency of a desired unmodulated 50 microvolt R.F. signal, and the frequency of a simultaneous undesired 0.5 volt R.F. signal, modulated 30 percent at 1000 Hz producing an interference output of 0.2 milliwatt at 1000 Hz, shall not be greater than 20 kHz.

3.15.2.7 Spurious responses, ant (See 4.9.2.7)- There shall be no response to an out-of-band frequency within +80 db of an on-tune Standard Test Signal.

3.15.2.8 Meter, tuning, ant (See 4.9.2.8)- An increase of R.F. input shall cause an increase in deflection. A 5 microvolt input signal of 200 kHz or higher, or a 10 microvolt input signal from 100 kHz up to, but not including 200 kHz shall cause discernible deflection from the no-signal position. A 100,000 microvolt input signal shall produce a deflection of at least four scale divisions and the pointer shall not go against the stop. With the "AUDIO" control adjusted for 1.5 milliwatts output and with R.F. inputs from 10 to 100,000 microvolts, the tuning pointer shall deflect discernible from its off-resonance position under no-signal condition as the ADF is tuned through resonance.

3.15.2.9 Resonance stability, ant (See 4.9.2.9)- With an unmodulated R.F. signal within the frequency range of 100 kHz to 3000 kHz and of any level up to 0.5 volts on the sense antenna input, there shall be no evidence of unstable operation, frequency modulation, flutter or excessive noise for any resonance frequency.

3.15.2.10 Calibration accuracy, ant (See 4.9.2.10)- With any combination of on-tune conditions the "Kilocycles" dial shall indicate the frequency within ±0.5 of the measured 6 db bandwidth.

3.15.2.11 Audio Fidelity, ant (See 4.9.2.11)- The fidelity of the audio circuits of this equipment shall be within the limits stated below.

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Audio-frequency Hz	db from 400 Hz Level	
	Lower Limit	Upper Limit
200	-15	0
300-1000	-3	+3
1500	-6	+3
2000	-14	+3
2500	-25	+2
3500	More than 25 db down	0
7000	More than 25 db down	-20

3.15.2.12 Audio Control, ant (See 4.9.2.12)-

3.15.2.12.1 R.F. input/audio output (See 4.9.2.12.1)- With any setting of the "AUDIO" control there shall be an approximately linear relationship between the r.f. input at any percentage modulation and the audio output in decibels relative to 1 milliwatt over the range minus 3 db to plus 15 db. With the "AUDIO" control set to give between 1 and 50 milliwatts at any r.f. level between 1 microvolt and 0.1 volt, the audio output shall increase as the r.f. level is increased as specified.

3.15.2.12.2 Rotation/audio output (See 4.9.2.12.2)- With r.f. input signals up to 100,000 microvolts and of any depth of modulation up to 80 percent, at least 7 degrees rotation of the AUDIO control shall be required to reduce the audio level from 16 milliwatts to 1.5 milliwatts. With any r.f. input up to 0.1 volt and of any depth of modulation up to 80 percent, the AUDIO control shall be capable of reducing the audio output to less than 100 microwatts.

3.15.2.12.3 Distortion harmonic (See 4.9.2.12.3)- With r.f. inputs up to 0.1 volt and with any depth of modulation between 30 and 80 percent and at any audio-frequency between 400 and 2000 Hz, the percentage harmonic distortion shall not exceed 20 percent when the AUDIO control is adjusted to give from 1 to a maximum of 32 milliwatts at the output.

3.15.3 "Loop" mode (See 4.9.3)-

3.15.3.1 Sensitivity, loop (See 4.9.3.1)- The loop sensitivity as measured at the audio output, when receiving the signals modulated 30 percent at 1000 Hz, and the "AUDIO" control being set to give 50 milliwatts output, shall be less than 12.5 milliwatts when modulation is removed. The audio output shall be 50 mw at each of the following frequency bands:

Frequency(kHz)	Field (uV/M)
100	360
200	250
400	200
800 and above	175

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(The loop sensitivity of this equipment at any frequency within its calibrated range under any combination of service conditions shall not degrade more than 6 db from the normal values listed).

3.15.3.2 Sensitivity CW, loop (See 4.9.3.2)- The "CW" sensitivity for a 10 milliwatt audio output shall be at least 6 db better than that specified for a 30%, 1000 Hz, modulated signal.

3.15.3.3 Rotation speed, loop (See 4.9.3.3)- When using the "L-LOOP-R" switch, the speed of the loop rotation shall be continuously variable and have a speed of 25 degrees +5 -0 per second.

3.15.3.4 Switch, reversing, loop (See 4.9.3.4)- A loop reversing switch will be internally mounted in the Control C-7392( )ARN-89, to be used in conjunction with the "L-LOOP-R" front panel switch. This switch will be used to reverse the direction of rotation of the goniometer in the loop mode so that a remote aircraft indicator will be compatible with a top or bottom mounted loop Antenna AS-2108( )/ARN-89. Markings shall be provided to indicate switch positions.

3.15.3.5 Audio control, loop (See 4.9.3.5)- With the function switch in the loop mode the audio control requirements shall be the same as 3.15.2.12; except when 100,000 microvolts are specified. 50,000 microvolts per meter shall be used. Otherwise where microvolts are specified, microvolts per meter shall apply.

3.15.4 "Comp." mode (See 4.9.4)-

3.15.4.1 Sensitivity, comp (See 4.9.4.1)- The sensitivity of the ADF throughout its frequency range under different service conditions shall be such that maximum variation of five repeated bearings taken on 100 kHz, 1.0 MHz and 2.9 MHz with 25 microvolts per meter and read on the bearing indicator shall not exceed 2 degrees. The sensitivity throughout its frequency range under different service conditions shall be such that maximum variation of five repeated bearings on an R.F. signal of 75 microvolts per meter and read on the bearing indicator shall not exceed 2 degrees.

3.15.4.2 Sensitivity, absolute, comp. (See 4.9.4.2)- The absolute sensitivity of the equipment will be considered as that field strength in microvolts per meter below which the bearings become uncertain, ambiguous or varying from the on-course bearing by more than 2 degrees. The sensitivity shall be such that a change of bearing of 1 degree under normal conditions for any signal input greater than 100 microvolts per meter will produce a change in indication of 1 degree.

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3.15.4.3 Selectivity, comp. (See 4.9.4.3)- When the specified undesired signal is traveling at right angles to the desired 1.0MHz signal, the selectivity of the equipment shall be such that the degrees variation from on-course bearings caused by an undesired signal shall in no case exceed the following values:

Undesired Signal from Desired	Microvolts per meter	Ratio of Undesired to Desired	Degree Error Maximum	F2 -F1 Maximum
3 kHz	158	10 db	3	6.0 kHz
6 kHz	28,000	55 db	3	12.0 kHz
7 kHz	79,000	64 db	3	14.0 kHz

3.15.4.4 Response, comp (See 4.9.4.4)- All undesired responses shall be at least 60 db down from the desired signal with a maximum bearing error of 2 degrees. The undesired signals shall be from 50 kHz to 3050 kHz at 50,000 microvolts per meter. Responses within a passband of 13 kHz about the center frequency shall be ignored.

3.15.4.5 Bearing, speed, comp. (See 4.9.4.5)- The equipment shall take a bearing of an R.F. signal at an average speed of at least 25 degrees per second under normal conditions and of at least 12 degrees per second under different service conditions from any starting point between 10 and 175 degrees from the bearing. The R.F. signal shall be of any frequency within the frequency range of this equipment of any field strength, between 100 and 50,000 microvolts per meter with modulation simulating broadcast radio range signals of the simultaneous type, or unmodulated continuous wave signals. Under normal conditions and with the above field strengths, the speed of taking bearings shall be at least 12 degrees per second when a continuous wave signal is received and is interrupted three times per second for a maximum of 1/6 of a second.

3.15.4.6 Lag, comp (See 4.9.4.6)- As the equipment is being rotated at any speed up to 10 degrees per second and taking a bearing on an R.F. signal, the indicated bearing shall not lag on the true on-course bearing by more than 5 degrees. As the equipment is being rotated at any speed between 10 and 25 degrees per second and taking a bearing on an R.F. signal, the indicated bearing shall not lag on the on-course by more than 10 degrees. The R.F. signal shall be of any frequency within the frequency range of this equipment, and of any field strength between 100 and 50,000 microvolts per meter with modulation simulating broadcast or radio range signals of the simultaneous type, or unmodulated continuous wave signals.

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3.15.4.7 Damping, comp. (See 4.9.4.7)- When the equipment is operating under normal or any service condition and the starting point is at any position up to 175 degrees from the true bearing, the overshoot of the indicator pointer shall not be greater than 2 degrees unless the overshoot and return occurs faster than can be noticed at a distance of three feet under normal illumination. Continuous hunting or oscillation of the pointer shall not exceed ±0.5 degrees. The pointer action shall be as dead-beat as is practicable and in no case shall there be more than one overshoot. The R.F. signal shall be any frequency within the frequency range of this equipment, of any field strength between 100 and 50,000 microvolts per meter with modulation simulating broadcast or radio range signals of the simultaneous type, or unmodulated continuous wave signals.

3.15.4.8 Accuracy, comp (See 4.9.4.8)- The degree variation of the on-course bearing obtained under normal conditions from the bearings for different conditions stated below shall not exceed 3 degrees:

a. Primary supply voltage variation within MIL-STD-704 limits.
b. Audio load impedance variation. (150 ohms ± 10%).
c. Sense antenna change with equivalent characteristics.
d. "AUDIO" Control variation.
e. Voice or keyed signals as described in 3.15.4.5.
f. Interchange of indicators, loops and loop cables with electrically equivalent units.
g. Tuning variation within R.F. passband of the desired signal.

3.15.4.9 Deviation corrector, comp. (See 4.9.4.9)- The deviation corrector shall be capable of compensating for errors ± 20 degrees, provided that the rate of change of error does not exceed 12 degrees in 15 degrees of azimuth, at every 30 degrees starting at 15 degrees for the complete 360 degree azimuth.

3.15.4.10 Headset, noise level, comp. (See 4.9.4.10)- When the goniometer is nulled to the bearing taken on unmodulated signals of any frequency within its calibrated range and of any field strength down to 200 microvolts per meter, the noise level in milliwatts at the headset shall not exceed 25 percent of the output obtained with 30 percent 1000 Hz modulation.

3.15.4.11 Audio control, comp. (See 4.9.4.11)- With any R.F. input up to 50,000 microvolts per meter and modulated up to 80 percent, the "AUDIO" control shall be capable of adjusting the audio output from the maximum obtainable to less than 25 microwatts. For any R.F. input up to 50,000 microvolts per meter and modulated up to 80 percent, there shall be an approximately linear relationship between degrees rotation and audio output in decibels with reference to 1 milliwatt, over the range - 3 db to 15 db. For any setting of the "AUDIO" control, the audio control action above 100 microvolts per meter shall be such that there is not more than a 10 to 1 variation in power output as the R.F. input is increased to 50,000 microvolts per meter.

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3.15.4.12 Distortion harmonic, comp. (See 4.9.4.12)- With any R.F. input up to 50,000 microvolts per meter and at any depth of modulation between 30 and 80 percent at any audio frequency between 400 and 2000 Hz, the percentage harmonic distortion shall not exceed 20 percent when the "AUDIO" control is adjusted to produce any output between 1.0 and a maximum of 32 milliwatts at the output.

3.15.4.13 Meter, tuning, comp. (See 4.9.4.13)- The action of the tuning meter shall be independent of variation of the "AUDIO" control. The maximum audio output due to modulation and the maximum deflection of the tuning meter pointer for voice modulated signals or keyed signals of simultaneous or non-simultaneous radio ranges shall occur simultaneously as the ADF is tuned through resonance. Increasing the R.F. input shall cause an increase in deflection. A 50 microvolt per meter R.F. input signal shall cause at least a 1/2 scale division deflection, and a 50,000 microvolt per meter R.F. input signal shall produce a deflection of at least four scale divisions and the pointer shall not go against the stop.

3.15.4.14 Filtering, sense and loop preamplifiers (See 4.9.4.14)- Filtering shall be provided at the inputs of the sense AM-4859( )/ARN-89 and loop AS-2108( )/ARN-89 preamplifiers. The filters shall provide over the frequency range of 30 to 400 MHz a minimum attenuation of 26 db in the loop preamplifiers and 34 db in the sense preamplifier. The phase difference between the loop and sense signals with the filters installed shall not exceed $\pm$ 33° over the 100 KHz to 3.0 MHz band.

3.15.5 Transient & reversal voltage (See 4.9.5)- The equipment shall not show any visual damage or degradation in the requirements of 3.15.2.1 and 3.15.4.8 after being subjected to a 80 volt transient and 27.5 volt reverse polarity applied to the power supply input.

3.15.6 Maximum R.F. input (See 4.9.6)- Regardless of any mode positions, the sense antenna input shall withstand R.F. voltages up to 100 volts rms and d.c. voltage up to a maximum of 50 volts without degradation.

3.16 Electromagnetic interference (See 4.10)- The unit(s) shall comply with the requirements of MIL-STD-826, except as otherwise indicated herein.

3.16.1 Method 5003, intermodulation, two-signal (See 4.10.1)- The unit(s) shall comply with Method 5003 as indicated herein.

3.16.2 Method 5001, Conducted susceptibility (See 4.10.2)- The unit(s) shall comply with Method 5001, except that the levels of figure 5 of MIL-STD-704 shall apply.

3.17 Weight (See 4.14)- The combined weight of the Directional Finder AN/ARN-89( ) shall not exceed 13.0 lbs.

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3.18 Burn-in (debugging) (See 4.6)- Each unit on order shall be capable of meeting the requirements herein, without subsequent processing after subjection to the Burn-in (debugging) as specified.

3.19 Operational inspection (See 4.12)- Each unit on order shall be subjected to the operational inspection as specified, after applicable processes, inspection and tests are completed, to determine that each unit is operable and set for the following normal conditions prior to packaging:

a. Loop reversal switch set to "Bottom" position.

b. Goniometer adjusted for zero (0) compensation.

c. Indicator selector set for single indicator.

3.20 Interchangeability (See 4.11)- Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the equipment.

3.21 Technical literature and spare parts - Technical literature and spare parts shall be furnished as specified in the contract. Spare parts furnished shall be identical to corresponding parts in the equipment furnished on order. (See 6.2)

3.22 Workmanship - The unit(s) shall be manufactured and assembled in accordance with the equipment drawings (See 3.2) and applicable portions of the following requirements and paragraphs herein:

MIL-STD-454:

Soldering	Requirement 5
Workmanship	Requirement 9

Paragraphs herein:

Cleaning	3.4
Control	3.5
Securing of parts	3.7
Soldering	3.9
Wiring & Cabling	3.10
Tropicalization	3.13

4. QUALITY ASSURANCE PROVISION

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4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.15 & 4.16).

4.3 Preproduction inspection - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (see Tables I and II and III respectively). The preproduction inspection will normally be performed in this order: Sample 1: (1) Temperature/altitude; (2) Humidity; (3) Vibration; and (4) Sand and dust; Sample 2: (1) Vibration (2) Temperature/altitude; (3) Humidity; (4) Shock, Bench handling and Equipment; and (5) Salt fog; Sample 3: (1) Electromagnetic interference; and (2) Explosion proof; Sample 4: (1) Acceleration; and (2) Fungus; other preproduction inspections may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Finish	3.6
Marking	3.8
Printed Wiring Assemblies	3.11
Semiconductors (Transistor & Diode)	3.12

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4.5 Quality conformance inspection of equipment before preparation for delivery - Each unit of each lot of equipment shall be subjected to Burn-in (debugging) (3.18) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of Burn-in (debugging) shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government - - at its discretion - - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.9) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105, using the general inspection levels and the AQL's indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.12) shall be last. (See 6.4)

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Table I - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.22	4.13		
Modules:				
Receivers:				
Mixer/I.F.			0.65%	2.5%
Audio/Servo			0.65%	2.5%
Power Supply			0.65%	2.5%
Goniometer:			0.65%	2.5%
Goniometer Assy.			0.65%	2.5%
Electronic Component Assy, Modulator			0.65%	2.5%
Electronic Component Assy, Splitter Bd.			0.65%	2.5%
Chassis, Wired, Assy.				
Control:				
Variable Osc.			0.65%	2.5%
Mixer Synthesizer			0.65%	2.5%
100 KHz Osc.			0.65%	2.5%
1.0 MHz Stepped Osc.			0.65%	2.5%
Chassis, Wired, Assy			0.65%	2.5%
Amplifier AM-4859( )/ARN-89 (less cover)			1.0%	4.0%
Antenna AS-2108( )/ARN-89 (less cover)			1.0%	4.0%
Control C-7392( )/ARN-89 (less cover)			1.0%	4.0%
Receiver R-1496( )/ARN-89 (less cover)			1.0%	4.0%
Direction Finder AN/ARN-89 ( ) (assembled)			1.0%	4.0%
Electrical:				
Sensitivity, ant.	3.15.2.1	4.9.2.1		
Sensitivity, CW, ant.	3.15.2.2	4.9.2.2		*1.0% for the combined group
Calibration accuracy, ant	3.15.2.10	4.9.2.10		
Sensitivity, loop	3.15.3.1	4.9.3.1		
Sensitivity, CW loop	3.15.3.2	4.9.3.2		
Rotation speed, loop	3.15.3.3	4.9.3.3		
Sensitivity, comp	3.15.4.1	4.9.4.1		
Sensitivity, absolute, comp	3.15.4.2	4.9.4.2		
Bearing speed, comp.	3.15.4.5	4.9.4.5		
Meter, tuning, comp.	3.15.4.13	4.9.4.13		
Operational inspection	3.19	4.12	0.1%	

*All electrical defects are considered major.

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4.5.2 Group B inspection - This inspection, including sampling, shall conform to Table II and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Order of inspection within Group B - Group B inspection shall be performed in any order which is satsifactory to the Government.

Table II - Group B Inspection -

Inspection	Req. Para.	Insp. Para.	AQL
Power Supply, primary	3.15.1.1	4.9.1.1	
Power Supply, 400 Hz	3.15.1.2	4.9.1.2	
R.F. Input/audio output	3.15.2.12.1	4.9.2.12.1	
Rotation/audio output	3.15.2.12.2	4.9.2.12.2	6.5% for
Distortion harmonic	3.15.2.12.3	4.9.2.12.3	the combined
Switch, reversing, loop	3.15.3.4	4.9.3.4	group
Audio control, loop	3.15.3.5	4.9.3.5	
I.F. rejection ant	3.15.2.3	4.9.2.3	
Image rejection, ant	3.15.2.4	4.9.2.4	
Selectivity, ant	3.15.2.5	4.9.2.5	6.5% for
Meter, tuning, ant	3.15.2.8	4.9.2.8	the combined
Audio fidelity, ant	3.15.2.11	4.9.2.11	group
Selectivity, comp	3.15.4.3	4.9.4.3	
Lag, comp	3.15.4.6	4.9.4.6	
Damping, comp	3.15.4.7	4.9.4.7	
Accuracy, comp	3.15.4.8	4.9.4.8	6.5% for
Deviation corrector,comp	3.15.4.9	4.9.4.9	the combined
Headset, noise level,comp	3.15.4.10	4.9.4.10	group
Audio control, comp	3.15.4.11	4.9.4.11	
Distortion harmonic,comp	3.15.4.12	4.9.4.12	
Interchangeability (Mech)	3.20	4.11	6.5%
Interchangeability (Elect)	3.20	4.11	6.5%

4.5.3 Group C inspection- This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.3.1.

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Table III - Group C Inspection -

Inspection	Rqt Para	Insp Para
Subgroup I:		
Temperature	3.14.1 thru 3.14.1.2	4.8.1 thru 4.8.1.2
Altitude	3.14.2 thru 3.14.2.1	4.8.2 thru 4.8.2.1
Cross modulation, ant	3.15.2.6	4.9.2.6
Spurious response, ant	3.15.2.7	4.9.2.7
Resonance stability, ant	3.15.2.9	4.9.2.9
Response, comp	3.15.4.4	4.9.4.4
Transient & reversal voltage	3.15.5	4.9.5
Maximum RF input	3.15.6	4.9.6
Subgroup II:		
Relative humidity	3.14.3	4.8.3
Vibration	3.14.5	4.8.5
Sand and dust	3.14.4	4.8.4
Bench handling	3.14.6.1	4.8.6.1
Equipment	3.14.6.2	4.8.6.2
Subgroup III:		
Fungus	3.14.7	4.8.7
Salt fog	3.14.8	4.8.8
Explosive proof	3.14.9	4.8.9
Acceleration	3.14.10	4.8.10
Warm-up equipment	3.14.11	4.8.11
Filtering, sense and loop preamplifiers	3.15.4.14	4.9.4.14
Electromagnetic interference	3.16	4.10
Weight	3.17	4.14
Subgroup IV:		
Reliability	3.14.2	4.8.12

4.5.3.1 Sampling for inspection- Unit(s) shall be selected at random for each group of inspections in Table III as specified in 4.5.3.1.1, through 4.5.3.1.4. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Selection of the samples shall be as specified and within 45 calendar days (within 60 calendar days for Subgroup IV) after selection of the samples have been subjected to and completed Group C inspections. (See 6.3).

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4.5.3.1.1 Subgroup I. - For this subgroup, two unit(s) from the first forty (40) production unit(s) shall be selected for the first Group C inspection. For subsequent Group C inspection, two unit(s) from each successive 100 or two per month if less than 100 or fractions thereof are produced monthly shall be selected.

4.5.3.1.2 Subgroup II. - For this subgroup, two unit(s) (these unit(s) may be the unit(s) selected for above subgroup 1), from the first forty (40) production unit(s) shall be selected for the first Group C inspection. For subsequent Group C inspection, two unit(s) from each 200 or fraction thereof or two every three months if less than 200 or fraction thereof are produced during three (3) successive months, shall be selected.

4.5.3.1.3 Subgroup III. - For this subgroup, two unit(s) from the first forty (40) production unit(s) shall be selected for the first Group C inspection. For subsequent Group C inspection, two unit(s) shall be selected every four (4) months except that fungus and salt fog shall be selected when one-half of the units on order have been accepted by the Government QA representative.

4.5.3.1.4 Subgroup IV. - For this subgroup, a minimum sample of nine (9) unit(s) for the Reliability Qualification (Demonstration) phase test shall be selected from the first twenty (20) production unit(s). For subsequent Group C inspections, a minimum sample of nine (9) unit(s) for the production acceptance tests, shall be selected from every two (2) months production; the samples shall be selected from different lots so that the unit(s) will be representative of the two (2) months production.

4.5.4 Non-compliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. For additional non-compliance see 4.8.12.6.

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4.5.5 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-in (debugging). - Each unit(s) on order shall be subject to a 48 hour burn-in period at test level E of MIL-STD-781, except that the duration of the vibration shall be at least 50 minutes of each on-time hour. The duty cycle shall be continuous and the temperature cycle shall not be greater than four (4) hours. The electrical parameters to be measured are those used for the Reliability Test (4.8.12.4). Measurements shall be taken during the first, one of the middle, and last temperature cycles. Measurements will be taken and recorded at both cold temperature and hot temperature during each measurement period. Measurements shall be taken during periods of vibration and no-vibration. The test records should indicate the environment at each reading (e.g. temperature + vibration). Measurements taken at cold temperatures shall start at -65 F and need only be made at one tuned frequency. If measurements taken during a period of no-vibration are not completed within ten minutes, the vibration shall be switched off until the measurements are completed. Each unit shall also be monitored to determine if intermittencies exist. An intermittency of one millisecond or greater duration will be considered to be a failure. Each unit which fails during the burn-in period must be repaired and returned to test until it successfully survives a 48 hour period without failure. Failures which occur during the burn-in will be noted and reported, but will not be counted towards the establishment of the MTBF. Prior to use on contract, the burn-in fixture, test procedures and temperature cycle shall be approved by the procuring activity. The requirements of 3.18 shall be met.

4.7 Standard test conditions. - Unless otherwise specified inspections shall be performed at room temperature, of 77 ±10°F and humidity not to exceed 90% under the following conditions:

a. Power supply voltage - 27.5 volts dc ± 0.5 volt.

b. Interconnecting R.F. cables: 25 ft.

c. Output load impedance (audio) - 150 ohms

d. Sense Antenna (simulating): 50 picofarad capacitor.

e. AM-4859( )/ARN-89, set for maximum gain, where a gain of minus 3 db is required, the following applies:

(1) Set the "OFF-COMP-ANT-LOOP" switch to "ANT".

(2) Set the R.F. input to 1 MHz, 50 microvolts, modulated 30 percent at 1000 Hz.

(3) Adjust the gain control of AM-4859( )/ARN-89 fully clockwise.

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(4) Adjust the "AUDIO" control until the tuning meter reads three divisions.

(5) Increase the R.F. input to 500 microvolts, and adjust the gain control of AM-4859( )/ARN-89 until the tuning meter again reads three divisions.

f. "Ant" mode tests. - Selector switch (S2) "OFF-COMP-ANT-LOOP" in "ANT" position and toggle switch (S1) "CW-VOICE-TEST" in "VOICE" position. All signal inputs refer to the signal on the generator side of a series 50 micro-microfarad capacitor (simulation the sense antenna), feeding the Amplifier, Impedance Matching AM-4859( )/ARN-89.

g. "Loop" mode tests. - Selector switch (S2) "OFF-COMP-ANT-LOOP" switch in "LOOP" position and toggle switch (S1) "CW-VOICE-TEST" in "VOICE" position. All signal inputs refer to the field strengths in the plane of Antenna AS-2108( )/ARN-89. The goniometer shall be rotated for maximum output.

h. "Comp" mode tests. - Toggle switch (S1) "CW-VOICE-TEST" in the "VOICE" position and selector switch (S2) "OFF-COMP-ANT-LOOP" switch in the "COMP" position. Signal inputs refer to field strengths in the plane of Antenna AS-2108( )/ARN-89. The gain of AM-4859( )/ARN-89 shall be set at maximum, (see 4.7.d.) and the sense input in microvolts shall be 0.05 times the loop input in microvolts per meter. (The sense input refers to the signal on the generator side of a series 50 uuF capacitor, simulating the sense antenna, feeding the Amplifier, Impedance Matching AM-4859( )/ARN-89).

4.7.1 Standard test signal. - The standard test signal shall be an amplitude modulated carrier, modulated to a depth of 30 percent at 1000 Hz, of such an amplitude that it produces a 6 db signal-plus-noise-to-noise ratio (reference to a 50 milliwatt signal-plus-noise-to-noise output) at the audio output terminals

4.8 Service conditions. - The service condition tests for the equipment shall be conducted in accordance with Specification MIL-T-5422 for Class 1 equipment as defined in 3.14. Electrical tests shall apply as specified in Table IV. No defects will be allowed.

4.8.1 Temperature. -

4.8.1.1 Operating. - Equipment shall meet the operating tests of MIL-T-5422 for compliance with 3.14.1.1.

4.8.1.2 Non-operating. - Equipment shall meet the non-operating tests of MIL-T-5422 for compliance with 3.14.1.2.

4.8.2 Altitude.- Equipment shall meet the operating and non-operating tests of MIL-T-5422 for compliance with 3.14.2.

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TABLE IV

		VIB		TEMP ALT.		SHOCK		ACCEL		HUMID		SAND & DUST		EXPLO-SION	
Requirements Para.	Test Freq KHz	Before and After	During	Before and After	During	Before and After	During	Before and After	During	Before and After	During	Before and After	During	Before and After	During
3.15.2.1	100 3000 100 (BFO on)	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
3.15.3.1	900 3000	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
3.15.4.1	500	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
3.15.4.5	500	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
3.15.4.7	500	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
3.15.2.10	100 210 1320 1430 1540 2650 2760 2870 2980	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
	Func. Oper.	To be conducted during Acceleration and Explosion													
	Mechanical Visual Inspection & Operation of Controls (See 3.5)	To be conducted before and after each Service Conditions Test													

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4.8.2.1 Temperature - Altitude comp. - Equipment shall be subjected to and meet the operating and non-operating tests of MIL-T-5422 for compliance with 3.14.2.1.

4.8.3 Relative humidity. - The equipment shall be subjected to and meet the humidity test of Specification MIL-T-5422 for compliance with 3.14.3. The unit(s) shall meet the operating test at the completion of each cycle after the third cycle.

4.8.4 Sand and dust. - The equipment shall be subjected to the sand and dust test of Specification MIL-STD-810, Method 510/1 for compliance with 3.14.4.

4.8.5 Vibration. - The equipment shall be subjected to the vibration test of MIL-T-5422 for compliance with 3.14.5.

4.8.6 Shock. -

4.8.6.1 Benchhandling. - The equipment shall be assembled for servicing and placed in a suitable position for servicing on a solid two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis for a total of four drops.

b. Repeat a. above, with the assembly resting on other faces, until it has been dropped a total of four times on each face on which the assembly could be practically placed for servicing. Tests shall be performed before and after shock, to insure compliance with paragraphs 3.15.2.1 and 3.15.4.1 at supply voltage of 27.5 volts.

4.8.6.2 Equipment. - The unit(s) shall be subjected to the shock tests of Specification MIL-T-5422. These tests shall include the 30 G Crash Safety Test and after this test, the equipment shall comply with the tests in 4.8.6.1.b.

4.8.7 Fungus. - The unit(s) including accessories shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition and as specified for compliance with 3.14.7 and shall not be required to be operated.

4.8.8 Salt fog. - The unit(s) including accessories shall be exposed in its normal operating condition to salt fog in accordance with Method 509 of MIL-STD-810 for 48 hours. The equipment shall not be opened and shall not be required to be operated. The equipment shall meet the requirement of 3.14.8.

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4.8.9 Explosive proof. - Unit(s) shall be tested for explosion proof in accordance with MIL-T-5422 for compliance with 3.14.9.

4.8.10 Acceleration.- The unit(s) shall be tested in accordance with Method 513, paragraph 3.3.5 of MIL-STD-810. This test shall be made to determine compliance with 3.14.10.

4.8.11 Warm-up equipment. - Set up the unit(s) to check the audio output. Energize the unit(s). Full output shall be noted with 30 seconds. These outputs shall be measured to determine compliance with 3.14.11.

4.8.12 Reliability. -

4.8.12.1 Qualification and production test plan. - The reliability of the equipment during qualification (pre-production) and production shall be demonstrated in accordance with Test Plan III, paragraph 4.2.8.3 of MIL-STD-781, using a specified MTBF of 660 hours.

4.8.12.2 Reliability test conditions. - Reliability Testing will commence on samples of units taken from lots which have successfully completed the 48 hour burn-in and have met Group A and Group B inspection. Test Level E, paragraph 4.1.6 of MIL-STD-781, shall be used for both qualification and production testing (except that the rate of temperature change of the thermal medium shall not be less than 1 C/minute). The duty cycle for the Direction Finding Set AN/ARN-89( ) shall be continuous. The total test time per unit shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781, and the temperature cycle shall not be greater than eight (8) hours.

4.8.12.3 Data recording interval. - Data will be recorded on a daily basis, i.e. seven days per week. The first data point will be taken during the heating cycle, starting when the chamber temperature reaches -25°F. Since it is not good practice to open the chamber at temperatures below room ambient (due to moisture condensation on the equipment), only data on parameters which can be accurately measured without the necessity of opening the chamber to manipulate equipment controls will be recorded at the -25°F temperature. The remainder of the required daily measurements will be taken starting when the chamber temperature passes through +68°F and continues until all data is recorded. At the conclusion of each day's measurements, the equipment will be set at the frequency to be read at -25°F, in order that the data may be taken the next day without the necessity of opening the chamber to set the frequency. The data taken at -25°F will be a different frequency every day.

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4.8.12.3.1 Data recording during vibration. - Measurement of each parameter on each unit will be taken at least once during the course of the test while the unit is under 2.2g vibration. Since the minimum test time for an accept decision is 11 days, each parameter must be measured under vibration conditions within 11 days of test start. For simplicity and ease of record keeping, this can be accomplished by measuring all required parameters on one unit per day under vibration conditions. In the event the data measurements have not been completed on this unit when the vibration cycle is complete (10 minutes of every ON hour), the vibration will be manually switched back on until the measurements have been completed. A note will be added to the data sheet noting that these measurements were made under vibration conditions.

4.8.12.4 Electrical parameter requirements. - Electrical parameter measurements, as specified in paragraphs 3.15.2.1 (at 3 frequencies; low, middle and high end of the band), 3.15.3.1 (at 3 frequencies, low middle and high end of the band), 3.15.4.1 (at convenient frequencies) and 3.15.4.5 (at convenient frequencies) will be made in the most efficient order to minimize test time.

4.8.12.4.1 Operating time measurement. - An elapsed-time meter will be used to monitor the number of operating hours accumulated for each equipment. The elapsed time meter will be connected such that only equipment "ON" time will be recorded.

4.8.12.5 Failure determination. -

4.8.12.5.1 Failure definition. - A failure is defined as the inability of the equipment to perform its required function. For the purpose of this test, any variation of the daily measured parameters below the limits in 4.8.12.4 will constitute a failure.

4.8.12.5.2 Failure classification. - Each failure occurring during this test will be classified as relevant or non-relevant. Only relevant failures will be counted in formulating an accept-reject decision.

4.8.12.5.2.1 Non-relevant failure. - A failure will be classified as non-relevant when analysis reveals the failure to be due to (a) accidental damage, (b) improper test equipment adjustment, (c) failure of another part (dependent failure), (d) test installation error, (e) failure of test equipment of facility (e.g. Prime Power), and (f) failure of less than three lamps.

4.8.12.5.3 Failed units. - On the occasion of a failure during the test, entries will be made in the appropriate data logs, the failed unit will be removed from test and the following actions will be initiated: (a) the failure will be confirmed, (b) the failure will be isolated and the repair made, (c) failure analysis will be initiated, (d) repair effectiveness will be verified. The test on the remaining units will continue while the failed unit is being repaired.

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4.8.12.5.3.1 Failure confirmation. - Everything external to the failed unit(s) will be closely examined to determine if it is accountable for the failure. If nothing external is found, it shall be assumed that the failure of the unit(s) is confirmed.

4.8.12.5.3.2 Failure isolation and repair. - Failures will be isolated to the smallest replaceable assembly, subassembly, card or part. Repair of the failed equipment will be limited to replacement of the defective assembly, subassembly, card or part, with an identical burned-in item. The failed item will then undergo analysis to determine the cause of failure.

4.8.12.5.3.3 Failure analysis and corrective action. - All failures detected during the Reliability Tests will be reported and analyzed. Correction action or recommended design changes will be submitted to the Commodity Division specified in the contract. Design changes will be submitted to the Commodity Division for approval. When pattern failures are observed during any reliability test of either an accept or reject decision, the contractor will notify the procuring activity within 10 days of the recommended corrective action to be taken.

4.8.12.3.4 Verifying effectiveness of repair. - Upon return of a unit from the repair area to the test area, following repair or corrective action, the unit will be re-installed in the test chamber allowed to operate one day under the Reliability Test conditions, and the repaired parameter measurements taken. The repair will be considered effective if the measured parameters are within the limits specified in 4.8.12.4. In the event performance is not satisfactory, the unit will be removed and returned to the repair area for further troubleshooting and repair. Failures and operating time accumulated during this period will be recorded and reported but not used in determining compliance with the MTBF requirement. Paragraph 5.6 of MIL-STD-781 shall apply.

4.8.12.6 Reliability demonstration compliance/Non-compliance. -

4.8.12.6.1 Compliance. - When the total unit hours of test, together with the total count of relevant equipment failures, result in an accept decision in accordance with the accept-reject criteria, compliance will have been demonstrated and testing will be terminated. Each of the test sample units which were subjected to and passed reliability testing will be shipped per instructions of the contracting officer, after all damage which may have occurred is repaired and the units successfully pass the Group A inspection.

4.8.12.6.2 Non-compliance. - When the total unit hours of test, together with the total count of relevant equipment failures result in a reject decision, non-compliance will be indicated and further shipment of units will be discontinued.

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4.8.12.6.2.1 Action resulting from Non-compliance. - Immediately upon reaching a reject decision, the procuring activity will be notified of the occurrence and the Reliability Test will be terminated until corrective action can be implemented on the lot from which the rejected test samples were drawn and all subsequent lots. Upon implementation of the approved corrective action, new test samples from the same lot will be submitted for retest. The sample size submitted for this re-test may be the same (nine units) or greater than the rejected sample size. Resumption of shipment shall be withheld pending the results of the retest. When the retest results in an accept decision and approved corrective action has been taken, shipment of units will be resumed. When the retest results in a reject decision, shipment of units shall be withheld and action shall be taken as described in this paragraph.

4.8.12.7 Thermal survey. - At the beginning of the program, a thermal survey of the equipments under test will be performed on a one-time basis only. The following points will be monitored: (1) surface temperature of front panel, (2) surface temperature of top cover, and (3) two points of surface temperature on back of radio set.

4.8.12.8 Demonstration of reliability. - The procedures and tests required by the paragraphs of 4.8.12 shall be performed to determine compliance with 3.14.12.

4.9 Electrical tests.-

4.9.1 Power supply. -

4.9.1.1 Power supply, primary. - Set the equipment to each mode of operation, with 27.5 vdc applied, the maximum input current shall not exceed 1.3 amperes. In the "Loop" mode, the goniometer shall be rotated left and right, and the 400 Hz power supply output shall have an equivalent load of one (1) Indicator ID-1351( )/A or equal. The input current shall be measured in each of the operating modes to determine compliance with 3.15.1.1.

4.9.1.2 Power supply, 400 Hz. - Connect the equivalent load of two (2) bearing Indicators (ID-1351( )/A) or equal to the equipment and adjust the power supply internal setting for nominal output voltage. While varying the input voltage between 21 and 29 volts dc, measure the output voltage and frequency for compliance with 3.15.1.2.

4.9.2 "ANT" mode (See 4.7.f). -

4.9.2.1 Sensitivity, ant. - Apply 10 microvolts, 100 KHz, modulated 30 percent at 1000 Hz and adjust the "Audio" control for a 50 milliwatt output across 150 ohm load. Remove the modulation. Repeat these tests at 200 KHz, 1000 KHz and 2900 KHz with 5 microvolts input. Measure the audio output when modulation is removed at each frequency to determine compliance with 3.15.2.1.

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4.9.2.2 Sensitivity CW, ant. - Apply 3.2 microvolts RF input at 100 KHz unmodulated and set switch (S1) to "CW" position and adjust "AUDIO" control for 10 milliwatts output across a 150 ohms load, repeat tests at 200 KHz, 1000 KHz, and 2900 KHz with 1.6 microvolts input. Measure the output at each frequency and when the input signal is removed to determine compliance with 3.15.2.2.

4.9.2.3 I.F. rejection ant. - Set R.F. input to 3 microvolts at 1 MHz. Tune AN/ARN-89( ) to signal and adjust "AUDIO" control to give a convenient reading on the "TUNE" meter. Increase input to 30 millivolts at the intermediate frequency. The requirements of 3.15.2.3 shall be met.

4.9.2.4 Image rejection ant.- Set R.F. input to 3 microvolts at 1 MHz. Tune AN/ARN-89( ) to the signal and adjust "AUDIO" control to give a convenient reading on the "TUNE" meter. Increase input to 30 millivolts at the image frequency. The requirements of 3.15.2.4 shall be met.

4.9.2.5 Selectivity, ant. - With an input of 5 microvolts, at 1 MHz, measure the 6 db and the 60 db bandwidth of the equipment. Measurements shall be made to ensure compliance with the requirements of 4.15.2.5.

4.9.2.6 Cross-modulation, ant.- Cross-modulation shall be measured by Method 5007 of MIL-STD-826, except as specified. The desired signal shall be set at 1 MHz, and the undesired signal shall be swept from 900 kHz to 1100 kHz. The requirements of 3.15.2.6 shall be met.

4.9.2.7 Spurious response, ant. - Set R.F. input to the standard test signal (See 4.7.1) at 1 MHz. Adjust "AUDIO" control to give an output of 50 milliwatts. Remove input. Feed in a signal 60 db higher in level, and sweep it over the range 50 kHz to 5 MHz. There shall be no response within -20 db of that given by the standard test signal, except in the band 990 kHz to 1010 kHz. Responses due to signal generator harmonic shall be ignored. The requirements of 3.15.2.7 shall be met.

4.9.2.8 Meter, tuning, ant. - Set gain control of AM-4859( )/ARN-89 for -3 db R.F. input at 1 MHz and 100 millivolts. The tuning meter shall deflect at least four divisions, but should not go against the stop. At various R.F. inputs between zero and 100,000 microvolts modulated 30 percent at 1000 Hz, set the "AUDIO" control to 50 milliwatts output. The tuning meter deflection shall increase at each level with an increase in R.F. level. Set the R.F. input inturn to 100,000 microvolts and 50,000 microvolts per meter 30 percent modulation at 400 Hz. Adjust "AUDIO" control for 1.5 milliwatts output. Tune AN/ARN-89( ) through resonance. There shall be a discernible deflection of the tuning meter. Set the gain control of AM-4859( )/ARN-89 to maximum, using input level of 10 microvolts. The requirements of 3.15.2.8 shall be met.

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4.9.2.9 Resonance stability, ant. - Set gain control of AM-4859( )/ARN-89 for -3 db. Set R.F. input to 10 microvolts, at 100 kHz, modulated 30% at 1000 Hz. Adjust "AUDIO" control for an audio output of 10 milliwatts. Remove modulation. Gradually increase input to a maximum of 0.5 volts, while noting autio output. Repeat at 1 MHz and 2.9 MHz. These tests shall be made for compliance with 3.15.2.9.

4.9.2.10 Calibration accuracy, ant. - With an input of 100 uv, measure the calibration accuracy of the equipment at the following settings of the "KILOCYCLES" control and the input frequency. The requirements of 3.15.2.10 shall be met.

"KILOCYCLES" dial setting	Input frequency
0100	100 kHz
0210	210 kHz
0320	320 kHz
0430	430 kHz
0840	840 kHz
1550	1550 kHz
1660	1660 kHz
2780	2780 kHz
2890	2890 kHz
2999.9	3000 kHz

4.9.2.11 Audio fidelity, ant. - With an R.F. input of 100 microvolts at 1 MHz, modulated 30 percent, measure the audio frequency response of the equipment relative to a 400 Hz, 25 milliwatts output. The requirements of 3.15.2.11 shall be met.

4.9.2.12 Audio control, ant. -

4.9.2.12.1 R.F. input/audio output. - Set R.F. input to 1 MHz modulated 30 percent at 1000 Hz. Set gain of AM-4859( )/ARN-89 for -3 db. At 1000 microvolts input adjust audio output to 1 milliwatt. Measure the input required to give audio outputs in the range 0.5 to a maximum of 32 milliwatts. Repeat at 80 percent modulation and 20,000 microvolts input. Set the R.F. input to 1 MHz 30 percent modulated at 1000 Hz. Set gain of AM-4859( )/ARN-89 for -3 db. With various input levels between 10 and 100,000 microvolts, and the output set to 50 milliwatts, the audio output shall increase as the R.F. input is increased. The measurements and the relationship between audio output/R.F. input shall be made for compliance with 3.15.2.12.1.

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4.9.2.12.2 Rotation/audio output. - Set R.F. input level to 100,000 microvolts modulated 30 percent at 1000 Hz. Set frequency to 1 MHz. Adjust "AUDIO" control for an output of 16 milliwatts with gain of AM-4859( )/ARN-89 set for -3 db. Rotate "AUDIO" control until output is reduced to 1.5 milliwatts. The "AUDIO" control shall have rotated at least 7 degrees. Rotate "AUDIO" control counterclockwise to minimum. The audio output shall be less than 100 microwatts. Repeat the tests at R.F. input levels of 100,000, 1000, and 10,000 microvolts and 80 percent modulation. The requirements of 3.15.2.12.2 shall be met.

4.9.2.12.3 Distortion harmonic. - With input level set at 1000 to 100,000 microvolts 1 MHz 30% modulated. Measure audio distortion with the "AUDIO" control set to give 1 milliwatt output at frequencies of 400, 1000 and 2000 Hz. Measurements shall be repeated at output levels of 8 and 32 milliwatts by adjustment of the "AUDIO" control at audio frequencies of 400 and 1000 Hz. This procedure shall be repeated at a modulation level of 80%. These measurements shall be made to determine compliance with 3.15.12.3.

4.9.3 "Loop" mode (See 4.7g). -

4.9.3.1 Sensitivity, loop. - Set field to 360 microvolts per meter at 100 kHz modulated 30% at 1000 Hz. Tune AN/ARN-89( ) to the signal. Set "AUDIO" control to give an audio output of 50 milliwatts. Remove modulation. Measure the audio output at each frequency specified to determine compliance with 3.15.3.1.

4.9.3.2 Sensitivity C.W., loop. - Set field to 88 microvolts per meter at 800 kHz unmodulated. Adjust the "CW-VOICE-TEST" switch to "CW". Retune AN/ARN-89( ) for maximum audio output. Set "AUDIO" control for an audio output of 10 milliwatts. Remove input signal. Audio output shall be less than 2.5 milliwatts for compliance with 3.15.3.2. Repeat this test at 1600 kHz.

4.9.3.3 Rotation speed, loop.- Using the "L-LOOP-R" control, subjective tests shall be made to ensure compliance with 3.15.3.3.

4.9.3.4 Switch, reversing, loop. - The switch will be tested in conjunction with the "L-LOOP-R" switch for goniometer rotation in the same direction selected by the "L-LOOP-R" switch with the switch set to bottom position, and goniometer rotation in the opposite direction from that selected by the "L-LOOP-R" switch with the switch set to top position. These switch positions shall be tested to determine compliance with 3.15.3.4.

4.9.3.5 Audio control, loop. - The equipment shall be tested as specified in the paragraphs of 4.9.2.12, except where 100,000 microvolts are specified, 50,000 microvolts per meter shall be used. Utilizing this value and microvolts per meter as specified, verify compliance with 3.15.3.5.

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4.9.4 "Comp" mode (See 4.7.h). -

4.9.4.1 Sensitivity, comp.- Set generator to each R.F. frequency specified at a level to give a field strength of 25 microvolts per meter at the AS-2108( )/ARN-89, record the bearing indicator reading. Press the CW/Voice/Test switch to TEST position and after bearing indicator stops release the switch and note the position where the bearing indicator stops. Repeat this procedure five (5) times at each of the frequencies to determine compliance with 3.15.4.1.

4.9.4.2 Sensitivity, absolute, comp. - Set the R.F. input to 25 microvolts per meter at 1 MHz. Gradually reduce the input level until the bearing becomes uncertain or ambiguous by more than two degrees. This input shall be designated the absolute sensitivity. Set the R.F. input to 100 microvolts per meter, at 100 kHz. Rotate Antenna AS-2108( )/ARN-89 by one degree. Repeat measurements at 1 MHz and 2.9 MHz. These measurements shall be made to determine compliance with 3.15.4.2.

4.9.4.3 Selectivity, comp.- This test shall be conducted using two loop signals at right angles in the plane of the Antenna AS-2108( )/ARN-89. The desired signal, to which the equipment is tuned, shall be 50 microvolts per meter. Provisions shall be made for accurate measurement of the undesired frequency. Set the frequency of the undesired signal lower than that of the desired, and adjust the frequency until the bearing indicator varies from the on-course bearing by three degrees. Designate this frequency f1. Repeat with the undesired signal higher in frequency than the desired. Designate this frequency f2. The requirements of 3.15.4.3 shall be met.

4.9.4.4 Response, comp. - This test shall be conducted using two loop signals at right angles in the plane of the Antenna AS-2108( )/ARN-89. The desired signal, to which the equipment is tuned, shall be 50 microvolts per meter, at 1 MHz. Provision shall be made for accurate measurements of the frequency of the undesired signals. Tests shall be made to determine compliance with 3.15.4.4.

4.9.4.5 Bearing, speed, comp. - Set the R.F. input to 100 microvolts per meter at 1 MHz. The goniometer shall be rotated 175 degrees clockwise from the on-course bearing and the time required for it to return to the on-course bearing shall be measured. The test shall be repeated for 175 degrees counterclockwise rotation. These tests shall be repeated with the signal keyed as specified. The requirements of 3.15.4.5 shall be met.

4.9.4.6 Lag, comp.- Measure the system lag with the Antenna AS-2108( )/ARN-89 rotated at both 10 degrees per second and 25 degrees per second. Measurements shall be made in both the clockwise and counterclockwise directions to determine compliance with 3.15.4.6.

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4.9.4.7 Damping, comp. - Set R.F. input to 100 microvolts per meter, 100 kHz modulated 80 percent at 1000 Hz. Adjust "OFF-COMP-ANT-LOOP" switch to "LOOP". Using "L-LOOP-R" control rotate indicated bearing 175 degrees off the on-course bearing. Adjust the "OFF-COMP-ANT-LOOP" switch to "COMP", and observe the action of the bearing indicator. Repeat at 50,000 microvolts per meter and at 1 MHz, and 2.9 MHz. These tests shall be made to determine compliance with 3.15.4.7.

4.9.4.8 Accuracy, comp. - Set the R.F. input to 100 kHz at 100 microvolts per meter. Perform the variations as specified one at a time to determine compliance with 3.15.4.8.

4.9.4.9 Deviation corrector, comp. - Set the Antenna AS-2108( )/ARN-89 on a rotating test table, and adjust the signal field strength to 1000 microvolts per meter at 1 MHz. Tests shall be made to determine compliance with 3.15.4.9.

4.9.10 Headset, noise level, comp. - Set input to 50,000 microvolts per meter at 100 kHz, modulated 30 percent at 1000 Hz. Set "AUDIO" control to give an output of 50 milliwatts. Remove signal modulation, and measure the output. Repeat at 200 microvolts per meter. These tests shall be made to determine compliance with 3.15.4.10.

4.9.4.11 Audio control, comp. - Set the R.F. input to 50 microvolts per meter, 1 kHz, 30 percent modulated at 1000 Hz. Measure audio output with the "AUDIO" control adjusted for maximum and minimum. Repeat with 50,000 microvolts per meter and with 80 percent modulation. Set R.F. input to 1 MHz, 100 microvolts per meter, modulated 30 percent at 1000 Hz. Adjust "AUDIO" control for 1 milliwatt output. Increase R.F. input to 1000, 10,000 and 50,000 microvolts per meter, recording audio output which shall not vary by more than 10 to 1. These tests shall be made to determine compliance with 3.15.4.11.

4.9.4.12 Distortion harmonic, comp. - Set R.F. to 1 MHz, 1000 microvolts per meter, modulated 30 percent. Adjust "AUDIO" control for 1 milliwatt output. With modulation frequencies of 400, 1000 and 2000 Hz, measure the harmonic distortion at the autio output. Repeat with audio control set for outputs of 8 and 32 milliwatts, at 400 and 1000 Hz. Repeat with an input of 50,000 microvolts per meter, and at 80 percent modulation. The requirements of 3.15.4.12 shall be met.

4.9.4.13 Meter, tuning comp. - Set the R.F. input to 50 microvolts per meter, 1 MHz, 30 percent modulated at 1000 Hz. Note the deflection of the tuning meter when the equipment is tuned to the input. Repeat at 50,000 microvolts per meter. Set the R.F. input to 200 microvolts per meter at 100 kHz, 30 percent modulated at 400 Hz. Set the audio output to 50 milliwatts. Detune the AN/ARN-89( ) and tune through the signal. The tuning meter maximum shall coincide with the 400 Hz audio output maximum. Repeat at 2 MHz. Set the R.F. input to 50 microvolts per meter, 1 MHz. The tuning meter deflection shall increase if the R.F. input is increased. Repeat at various input levels up to 50,000 microvolts per meter. These tests shall be made to determine compliance with 3.15.4.13.

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4.9.4.14 Filtering, sense and loop preamplifiers - Set R.F. to 30 MHz, 50,000 microvolts per meter (see paragraph 4.7h) unmodulated and measure the input to the first stage of the loop, prior to final encapsulating of the loop, and sense preamps of both an unfiltered reference set of amps and the filtered amplifiers under test. Observe the difference in signal levels between filtered and unfiltered amplifiers. Repeat R.F. frequencies of 40, 50, 60, 70, 76, 115, 125, 135, 145, 225, 250, 300, 350 and 400 MHz. Set the R.F. generator to 100 kHz, 1000 microvolts per meter unmodulated and measure the phase difference between the loop and sense signals at the receiver combiner stage. Repeat for R.F. frequencies of 200, 400, 1000, 2000 and 3000 kHz. The measurements shall be taken to determine compliance with 3.15.4.14, (see paragraph 6.8).

4.9.5 Transient and reversal voltage - Apply 27.5 volts dc reversed polarity to the power supply input and set the selector switch (S2) to each of its positions. With the equipment operating apply an 80 volt transient for 225 milliseconds in accordance with MIL-STD-704. After subjecting to these conditions measurements and inspection shall be made to determine compliance with 3.15.5.

4.9.6 Maximum RF input - With the unit(s) in operating mode, apply the required signal levels for one minute. Remove the test signal and make sensitivity measurements to determine compliance with 3.15.6.

4.10 Electromagnetic interference - The equipment shall be tested in accordance with MIL-STD-826, for Class AM equipment, except as noted herein. (See 3.16)

4.10.1 Method 5003, intermodulation, two signal - Unit(s) which do not comply with Method 5003 signals, change signals from each signal generator to 60 db rather than 66 db and repeat tests.

4.10.2 MIL-STD-826, Test Method 5001 - For Method 5001 apply the signal levels given by Figure 5 of MIL-STD-704.

4.11 Interchangeability - The mechanical dimensions and electrical performance required by the mechanical gage drawings and electrical test fixture drawing listed below shall be gaged or measured to determine conformance to the physical and functional interchangeability requirements of 3.20. When a listed dimension or performance requirement is not within specified or design limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured prior to assembly into the units:

a. Mechanical (dimension) - The gage points covered by the drawings listed on the applicable gage lists as listed in the contract for this order.

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b. Electrical performance (modules)- Prior to assembly into the equipment, plug-in modules shall be tested to determine compliance with the performance requirements specified in the applicable test procedures and gage lists as listed in the contract for this order.

4.12 Operational inspection- The equipment shall be operated and shall operate correctly as an ADF system on signals of greater than 25 microvolts per meter, and 27.5 vdc input whether derived from distant transmitting stations within the equipment frequency range, or from locally generated signals. (Substitution of simulated use with units is permissible when operational conditions are accurately duplicated. The substituted fixture shall be approved by the procuring Contracting Officer.). The equipment shall be operated in all modes of operation and shall operate satisfactorily for compliance with 3.19.

4.13 Visual and mechanical inspection - The unit(s) shall be examined for defects listed in MIL-STD-252. Prior to assembly into the unit(s) printed circuit boards shall be examined for compliance with MIL-P-55110 and assemblies, modules and sections shall be examined for compliance with the equipment drawings and MIL-STD-252.. The requirements of 3.22 shall be met.

4.14 Weight - The units of the AN/ARN-89( ) shall be combined and weighed to determine compliance with 3.17.

4.15 Quality conformance inspection of preparation for delivery - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the packaging requirements of Section 5.

4.16 Rough handling test (preparation for delivery)- When rough handling test in accordance with Specification MIL-P-116 is required by the contract the operational inspection test 4.12, shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging of Direction Finder Set AN/ARN-89( ) shall be Level A or C as specified in the contract or order (See 6.2b).

5.1.1 Level A

5.1.1.1 Cleaning - Each direction finder set shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying - Each direction finder set shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application - None required.

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5.1.1.4 Unit packaging - Each direction finder set shall be packaged in accordance with items 1 through 8 of Table V and Figure 1.

5.1.2 Level C - Each direction finder set shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing- Packing of Direction Finder Set AN/ARN-89( ) shall be Level A, B or C as specified (See 6.2).

5.2.1 Level A -

5.2.1.1 Palletized load - A quantity of direction finder sets, packaged as specified in 5.1, shall be placed on a pallet, load Type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load - When quantities per destination are less than a pallet load, a quantity of items, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, Style E or MIL-B-43291. The gross weight of the boxes conforming to PPP-B-640 shall not exceed 200 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by non-metallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, Type II or PPP-T-97, Type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B - Each direction finder set, packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C - Each direction finder set, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Reinforcing of boxes, as specified in 5.2.1.2, will not be required. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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BILL OF MATERIAL
TABLE V

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Technical Literature	1	IC-1	11	8-1/2	1/2							
	Bag	1		13	10		MIL-B-117	II	C				
2	Direction Finder Set AN/ARN-89( )	1	IIb				(Bare wt. 13.5 lbs)						
	consisting of the following items stored without packaging within the case, Ref No. 5												
	Receiver, Radio R-1496 ( )/ ARN-89	1		11	6	5-1/2							
	Control, Radio C-7392( )/ ARN-89	1		6	5-3/4	3-3/4							
	Amplifier, AM-4859 ( )/ ARN-89	1		3-7/8	3-1/4	1-5/8							
	Antenna AS-2108( )/ ARN-89	1		11-7/8	11-7/8	1							
3	Cover, Case	1		13-1/8	9-1/2	6-5/8	PPP-C-850 (1-1/2 lbs. per cu. ft. density)						
4	Desiccant	3		5	3-1/2	5/8	MIL-D-3464 (4 Unit)	II					
5	Case	1		13-1/8	9-1/2	6-5/8	PPP-C-850 (1-1/2 lbs. per cu. ft. density)						
6	Humidity Indicator	1		3	2		MIL-I-8835						
7	Bag, Barrier	1		28 (Inside dimensions)	24		MIL-B-131		1				
8	Box, Fiberboard	1		13-5/8	10-1/16	14-1/8	PPP-B-636	CF	WR	SW	W5c		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

AMSEL-MR/P Form 3135
1 Oct 1965

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DIRECTION FINDER SET
AN/ARN-89( )

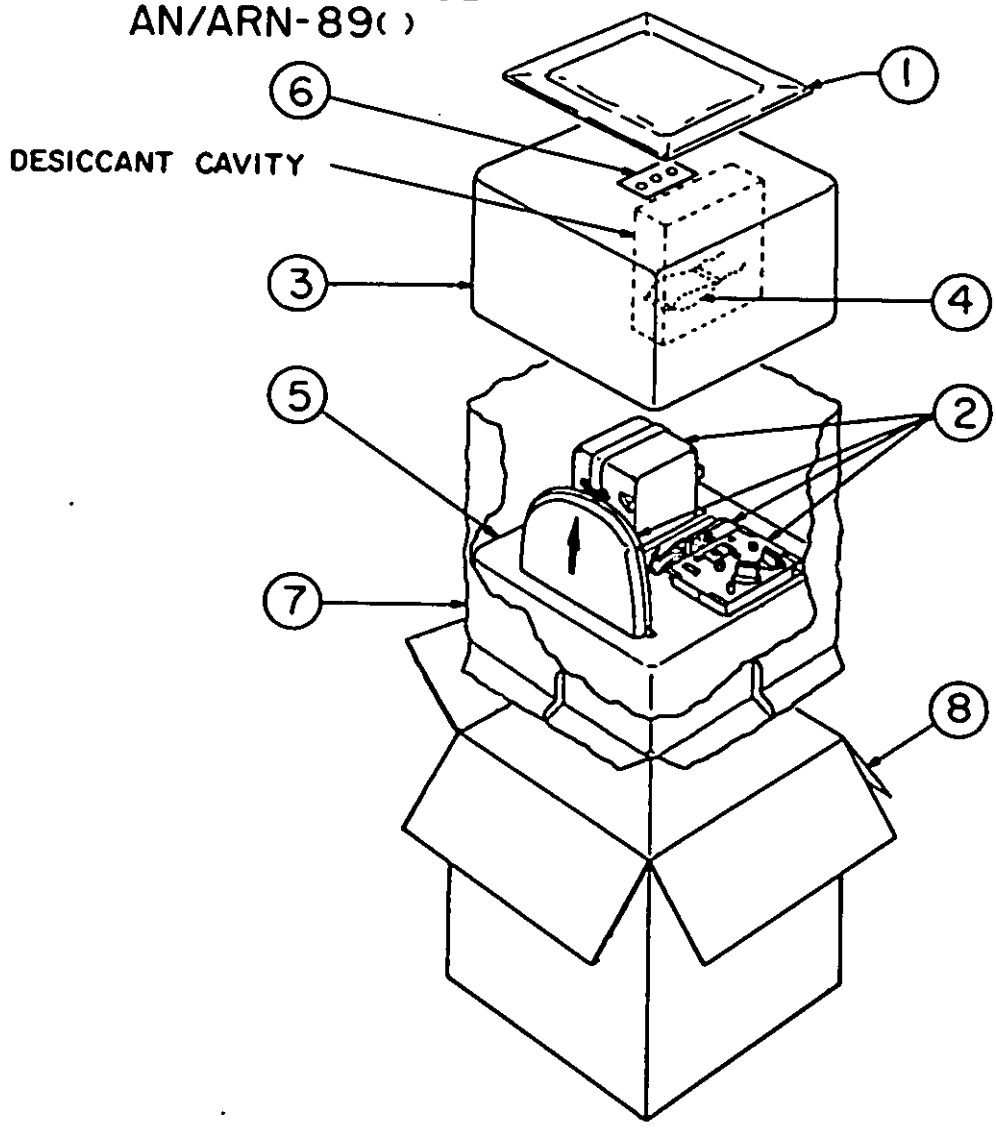


FIGURE 1

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6. NOTES

6.1 Intended use - Direction Finder Set AN/ARN-89( ) is intended for use by the pilot and copilot to provide manual or automatic directional information on any radio signal of sufficient signal strength within the frequency range of 100 kHz to 3000 kHz, and reception of communications over the same frequency range when installed in aircrafts.

6.2 Ordering data- Procurement documents should specify the following:

a. Title, number and date of this specification any any amendment thereto.

b. Levels of preservation, packaging and packing required. (Level A, Level B or Level C).

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. Preproduction inspection:

(1) Several sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit. (See 3.3)

(2) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon. (Including rough handling).

e. Marking and shipping of sample.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of spare parts required.

i. Submission of the statement of treatment referenced in 3.13 should be submitted as specified in the solicitation/contract. This statement should be submitted to the contracting officer.

6.3 Group C inspection - Approval to ship unit(s) on order may be withheld, at the discretion of the Government pending decision from the procuring activity, until compliance with 4.5.3.1 is complete.

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6.4 Location of operational inspection - It is desirable that the operational inspection 4.12 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.5 Nomenclature - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/ARN-89W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.6 Definition of Level B - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.7 Dimensional data - Dimensions of packaging materials in the Bill of Materials Table are based on the dimensions of the equipment cited in the table. When the dimensions of the items vary, the dimensions of the packaging materials shall be adjusted accordingly.

6.8 Filtering, sense and loop preamplifiers - The contractor shall record actual readings on the first 100 units of production and make the data available on request. After the testing of the first 100 each unsealed loops, the successful bidder (contractor) shall establish equivalent requirements and tests for the AS-2108( )/ARN-89 after assembly is completed to determine that the loop is capable of compliance with the requirements of 3.15.4.14. The tests, after approval by the Government (USAECOM), shall be used during the inspection specified for the 4.9.4.14.

6.9 Verification inspection - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian
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Preparing Activity
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Project No. 5826-A075

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